FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

CNS & Musculoskeletal Examination

Clinical skill | Week 2 · Diarrhoea, fluids, HIV & anaemia | Prof Karamagi, Dr Nabatte

The paediatric neurological and musculoskeletal examination is adapted to the child's age (observation and play first, hands-on last), always starts with ETAT emergency signs and conscious level (AVPU, Blantyre coma scale or modified GCS), and must cover head size, meningism, cranial nerves, tone, power, reflexes, coordination, gait, spine and joints (pGALS), ending with a classified summary that drives management.

Must know before the ward round

1. Start with ABCD and conscious level: AVPU (P or U = coma, emergency) then Blantyre coma scale (pre-verbal children) or modified/adult GCS; GCS ≤8 = airway at risk.
2. Blantyre coma scale (max 5): eyes 1 (watches/follows) or 0; best motor 2 (localises pain), 1 (withdraws), 0 (none/inappropriate); best verbal 2 (cries appropriately/speaks), 1 (moan/abnormal cry), 0 (none). ≤2 = unrousable coma (cerebral malaria); UCG: deep coma = unable to localise pain.
3. Observe before you touch: posture, spontaneous movements, asymmetry, involuntary movements, how the child sits, crawls, walks and uses the hands. Most of the infant neuro exam is done by watching and playing.
4. Always measure head circumference and plot it (micro- < −2 SD, macro- > +2 SD); feel the fontanelle with the child upright and calm (bulging = raised ICP/meningitis; sunken = dehydration).
5. Meningism: neck stiffness, Kernig, Brudzinski are unreliable under 12–18 months; in infants look for a bulging fontanelle, irritability, poor feeding, high-pitched cry.
6. Raised ICP signs (WHO; contraindicate LP): unequal or poorly reactive pupils, abnormal posturing, focal paralysis, irregular breathing, bradycardia with hypertension, papilloedema, deteriorating coma score.
7. UMN lesion: increased tone (spastic catch), brisk reflexes, clonus (>3–5 beats), upgoing plantar (abnormal after ~12–18 months). LMN: wasting, fasciculation, hypotonia, reduced/absent reflexes. Muscle: proximal weakness, Gowers sign, normal or reduced reflexes.
8. Reflex roots: biceps C5–6, supinator C5–6, triceps C7, knee L3–4, ankle S1. Grade 0 absent, 1+ reduced, 2+ normal, 3+ brisk, 4+ with clonus. MRC power 0–5.
9. Primitive reflexes (Moro, palmar grasp, ATNR) gone by 4–6 months; parachute appears 6–9 months. Persistence = CP; asymmetric Moro = brachial plexus injury, clavicle fracture, hemiplegia.
10. Acute flaccid paralysis in a child under 15 = possible polio: notify the district surveillance focal person, 2 stool samples 24 h apart within 14 days of onset, NO IM injections (UCG 2023).
11. pGALS (paediatric Gait, Arms, Legs, Spine) screens the MSK system in about 2 minutes; then 'look, feel, move' any abnormal joint. A hot, swollen, painful joint in a febrile child is septic arthritis until proven otherwise: aspirate and give cloxacillin 50 mg/kg IV 6-hourly (UCG).
12. Uganda specifics: fundoscopy for malarial retinopathy in coma; sciatic nerve injury from buttock injections (foot drop); Pott's spine (gibbus); sickle cell dactylitis and osteomyelitis (Salmonella); rickets; nodding syndrome in northern Uganda.

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1. Purpose and Indications

Examining a child's **nervous system and musculoskeletal system** is a skill that rewards **patience and play** more than a reflex hammer. In infants and toddlers, most of the examination is done by **watching**: how the child lies, moves, reaches, sits, crawls, walks, talks and interacts. Hands-on and unpleasant parts (fundoscopy, reflexes, painful joints, meningism) come **last**.

On a Ugandan paediatric ward, the CNS examination is used every day for **coma** (cerebral malaria, meningitis, hypoglycaemia, poisoning), **convulsions**, **weakness** (polio-like acute flaccid paralysis, Guillain–Barré syndrome, sickle cell stroke, injection injury), **developmental delay** and **cerebral palsy**. The MSK examination catches **septic arthritis, osteomyelitis, pyomyositis, rickets, TB spine, sickle cell crises and rheumatic fever**. The examination must end in a **summary and classification** (for example "Blantyre 2, no meningism, no focal signs" or "UMN pattern, spastic diplegia") that tells you what to do next.

> **CLINICAL PEARL**
> The one thing never to forget: **in a sick child, the neurological examination begins with ABCD and the conscious level, not with the reflex hammer.** A child who responds only to pain (P) or not at all (U) on AVPU is in coma: check glucose, stop fits, protect the airway and start treatment before you complete the rest.

Indications: any child with seizures, altered consciousness, headache, fever with irritability or drowsiness, weakness or abnormal gait, developmental delay or regression, abnormal head size, suspected CP, limp, joint or bone pain or swelling, back pain or deformity, and as part of the routine newborn check.

Key terms

Term	Meaning
AVPU	Alert, responds to Voice, responds to Pain, Unresponsive. P or U = coma (WHO ETAT)
Blantyre coma scale (BCS)	Coma scale for pre-verbal children (developed in Malawi for cerebral malaria). Score 0–5 (eyes 0–1, motor 0–2, verbal 0–2)
Glasgow coma scale (GCS), modified for children	3–15 (eyes 1–4, verbal 1–5, motor 1–6); verbal component adapted for infants
Upper motor neurone (UMN) lesion	Lesion of brain or spinal cord: spasticity, hyperreflexia, clonus, upgoing plantar, weakness in a pyramidal pattern
Lower motor neurone (LMN) lesion	Anterior horn cell, root, plexus or nerve: flaccid weakness, wasting, fasciculation, reduced or absent reflexes
Spasticity	Velocity-dependent increase in tone ("catch" on fast stretch; clasp-knife)
Rigidity	Increased tone throughout the range regardless of speed (lead-pipe); extrapyramidal or dystonia
Hypotonia	Reduced tone ("floppy"): central (brain: preserved or brisk reflexes, often poorly alert) or peripheral (neuromuscular: weakness, absent reflexes, alert face)
Ataxia	Incoordination (cerebellar, sensory)
Meningism	Signs of meningeal irritation: neck stiffness, Kernig, Brudzinski, photophobia

Term	Meaning
pGALS	Paediatric Gait, Arms, Legs, Spine: a validated MSK screening examination for school-age children (Foster and Jandial)

2. Contraindications and Precautions

There are no contraindications to the examination itself, only **precautions**:

- **Suspected cervical spine injury** (trauma): do not test neck stiffness or move the neck; immobilise.
- **Raised ICP**: do not perform an LP; minimise distress.
- **Unstable child**: stabilise first (ETAT), do the full examination later.
- **Painful joints**: examine the painful joint last and gently; never force movement.
- **Suspected septic joint or fracture**: avoid weight-bearing tests; immobilise.
- **Painful stimuli for coma scoring**: use the least painful stimulus that gives the answer (pencil on the nail bed or sternum); never needles or pinching that bruises.
- **Acute flaccid paralysis**: no IM injections (provocation polio).

First 5 minutes: is it safe to do a full examination now?

1. **ETAT**: airway, breathing (SpO_2, RR), circulation (capillary refill, pulse), **coma** (AVPU), **convulsions**, dehydration.
2. **Glucose** in any child with altered consciousness or seizures; treat < 3.0 mmol/L (UCG; WHO < 2.5, < 3.0 in SAM) with 10% dextrose 5 mL/kg (neonate 2 mL/kg).
3. **If convulsing**: treat first (diazepam 0.5 mg/kg PR, max 10 mg; see Febrile Convulsions).
4. **If trauma**: immobilise the cervical spine before any neck examination.
5. **If unstable**: do a **focused** neurological examination (coma score, pupils, posture, limb movements, fontanelle, neck stiffness unless trauma) and complete the rest later.

> UGANDA CONTEXT
> **Hypoglycaemia threshold**: UCG 2023 treats glucose **< 3.0 mmol/L**; the WHO Pocket Book uses **< 2.5 mmol/L** (< 3.0 mmol/L in severe acute malnutrition). On Ugandan wards use the UCG threshold.

3. Relevant Anatomy and Normal Values

Reflexes, roots and grading

Reflex	Root
Biceps	**C5–6**
Supinator	**C5–6**
Triceps	**C7**
Knee	**L3–4**
Ankle	**S1**

Reflex grade: 0 absent, 1+ reduced, 2+ normal, 3+ brisk, 4+ with clonus.

Myotomes for power testing (older child)

Movement	Root
Shoulder abduction	C5
Elbow flexion / extension	C5–6 / C7
Wrist extension	C6–7
Finger abduction	T1
Hip flexion	L1–2
Knee extension	L3–4
Ankle dorsiflexion	L4–5
Plantar flexion	S1–2

MRC power scale: 0 no contraction; 1 flicker; 2 movement with gravity eliminated; 3 against gravity; 4 against some resistance; 5 normal.

Modified Ashworth Scale (tone): 0 no increase; 1 or 1+ slight catch; 2 marked increase but easily moved; 3 considerable increase, passive movement difficult; 4 rigid.

Age-related normal values

Sign	Normal
Posture of a term newborn	Flexed
Fix and follow a face or bright toy	By **3 months**
Head lag on pull-to-sit	Abnormal after **4 months**
Primitive reflexes (Moro, palmar grasp, ATNR)	Gone by **4–6 months**
Parachute reflex	Appears **6–9 months**
Upgoing (extensor) plantar	Normal until about **12–18 months** (abnormal once walking is established)
Ankle clonus	A few beats can be normal in newborns; **> 3–5 beats sustained** is abnormal beyond that
Meningism (neck stiffness, Kernig, Brudzinski)	**Unreliable under 12–18 months**
Head circumference	Microcephaly < −2 SD; macrocephaly > +2 SD
Physiological genu varum (bow legs)	Normal to about **2 years**
Physiological genu valgum (knock knees)	Normal at **3–5 years**
Hip examination (Ortolani and Barlow)	Up to about **3 months**; then limited abduction, leg-length difference, asymmetric thigh creases, Galeazzi sign

Neurocutaneous stigmata

- **≥ 6 café-au-lait macules** > 5 mm prepubertal or > 15 mm postpubertal: neurofibromatosis type 1.
- **Ash-leaf macules and facial angiofibromas**: tuberous sclerosis.
- **Port-wine stain on the face**: Sturge–Weber syndrome.

4. Equipment and Preparation

Equipment

Item	Use
Tape measure (non-stretch)	Head circumference; limb lengths
Growth and HC chart (WHO)	Plot HC Z-score
Toys: bright rattle or ball, small cubes, a small bean or raisin (supervised), crayon and paper	Vision, reach, pincer grip, fine motor, coordination
Pen torch	Pupils, eye movements
Ophthalmoscope	Red reflex, fundi (papilloedema, retinal haemorrhages, **malarial retinopathy**)
Tendon (reflex) hammer	Deep tendon reflexes
Orange stick, cotton wool, tuning fork (128 Hz)	Sensation in older children; plantar response (use a blunt orange stick, not a needle)
Glucometer	Every child with altered consciousness or seizures
Prader orchidometer	Puberty (not part of the CNS examination, but often in the same visit)
Gloves, privacy screen, a parent present	Comfort, consent, chaperone

Age-appropriate approach

Age	Approach
Newborn and young infant (< 6 months)	Observe posture (flexed in a term baby), spontaneous movements, cry, suck; tone (pull-to-sit, ventral and vertical suspension); primitive reflexes; HC, fontanelles, sutures; spine and hips
Older infant and toddler (6 months–3 years)	Observe on the parent's lap; play to test reaching, pincer, transfer, sitting balance, standing, walking; tone and reflexes once the child is settled; parachute reflex
Preschool (3–5 years)	Make it a game: "walk like a soldier, on tiptoes, on heels, hop"; "touch my finger then your nose"; jump; draw shapes
School-age and adolescent	Near-adult examination (formal cranial nerves, MRC power, sensation, coordination), pGALS, Tanner staging where relevant; respect privacy

Focused history (what to ask before examining)

Ask	Why it matters
Onset and course	Sudden (stroke, trauma, haemorrhage); acute over hours to days (infection, Guillain–Barré, poisoning); subacute (TB meningitis, tumour); chronic static (CP); progressive (neurodegenerative, HIV, tumour, muscular dystrophy)
Fever, headache, vomiting, neck pain, photophobia, rash	Infection (meningitis, malaria), raised ICP
Seizures (description, duration, focal features) and **drugs given**	Post-ictal state, drug sedation, number of benzodiazepine doses; focal seizure with Todd's paresis
Weakness: where (one side, both legs, proximal, distal), ascending, bladder and bowel involvement, preceding diarrhoea or respiratory illness	Guillain–Barré syndrome, cord lesion, AFP

Ask	Why it matters
Injections into the buttock or thigh; **polio immunisation** status	Injection neuritis; polio
Joint or bone: pain, swelling, redness, limp, refusal to walk or move a limb, trauma, fever, migratory pattern, morning stiffness, night pain, **sickle cell disease**	Septic arthritis, osteomyelitis, fracture; rheumatic fever (migratory); JIA (morning stiffness); tumour (night pain); sickle crisis
Development and regression; **birth history**; **head injury**	CP, neurodegenerative disease, HIE, abusive head trauma
Immunisation (UNEPI: OPV, IPV, measles–rubella, Hib, PCV), **HIV status**, **TB contact**, **feeding and nutrition**	Polio, meningitis risk, HIV encephalopathy, TB meningitis or spine, rickets
Travel and context: northern Uganda; poisoning (organophosphates, paraffin, traditional herbs)	Nodding syndrome; toxic coma

Preparation

- **Consent** from the parent and **assent** from the child; **parent present** throughout.
- **Privacy**; expose to vest and nappy or shorts.
- **Warm, calm room**; toys ready; plan to do the **unpleasant parts last** (fundoscopy, reflexes, meningism, painful joints, pain stimulus for coma scoring).

5. Procedure Step by Step

The sequence follows Hutchison's Clinical Methods (24th edition), adapted for children: **observe first**, test by play, do hands-on and unpleasant parts last, and judge each sign against **age-specific normals** (section 3). In an emergency, ETAT and the conscious level come first (section 2).

A. Introduction, consent and preparation

1. **WIPER**: **W**ash hands; **I**ntroduce yourself to the parent and child (by name), explain, get **consent** from the parent and **assent** from the child; **P**rivacy; **E**xpose appropriately (vest and nappy or shorts); **R**eposition the child on the parent's lap or bed, **parent present**.
2. Keep the child **warm** and comfortable; the parent holds or stays with the child.

B. General inspection from the end of the bed

1. **ABCD**, conscious level, distress.
2. **Posture**, **spontaneous movements** (symmetry), **involuntary movements** (seizures, chorea, dystonia, tremor).
3. **Dysmorphic features**; **neurocutaneous stigmata** (section 3).
4. **Skin**: petechiae, purpura, bruises.
5. **Nutrition**: wasting, oedema.
6. **Aids**: wheelchair, splints, ankle–foot orthoses (AFOs), shunt reservoir behind the ear, gastrostomy.
7. **Vital signs and glucose** in an ill child: HR, RR, **BP** (Cushing response), temperature, SpO_2, glucose.
8. **Growth**: weight, length or height; head circumference (step D) plotted.

C. Conscious level

Blantyre coma scale (pre-verbal children)	Score
Best motor response: localises painful stimulus (firm pressure with the blunt end of a pencil on the sternum or supraorbital ridge)	2
Withdraws limb from pain (pressure on a nail bed with a pencil held horizontally)	1
No response or inappropriate response	0
Best verbal response: cries appropriately with pain (or speaks, if verbal)	2
Moan or abnormal cry with pain	1
No vocal response to pain	0
Eye movements: watches or follows (for example the mother's face)	1
Fails to watch or follow	0
Total	**0–5** (≤ 2 = unrousable coma; 5 = normal)

Modified GCS (child < 5 years)	Score
Eyes: spontaneously 4; to voice 3; to pain 2; none 1	1–4
Verbal: alert, babbles, coos or words as usual 5; fewer words than usual or irritable cry 4; cries only to pain 3; moans to pain 2; none 1	1–5
Motor: normal spontaneous movements or obeys 6; localises pain (or withdraws to touch in infants) 5; withdraws from pain 4; abnormal flexion (decorticate) 3; abnormal extension (decerebrate) 2; none 1	1–6
Total	3–15; **≤ 8 = protect the airway**

Always record the **components** (for example "BCS 2 = M1 V1 E0") and the **time**: coma scores change.

D. Head, face and neck (including meningism)

1. **Head circumference**: maximal occipitofrontal; plot; compare with previous measurements.
2. **Shape**: plagiocephaly, scaphocephaly (craniosynostosis), frontal bossing (rickets, hydrocephalus).
3. **Fontanelles and sutures** (child upright and calm): size, tension (bulging, flat, sunken), split sutures, overriding sutures (microcephaly).
4. **Signs of hydrocephalus**: scalp veins, sunsetting eyes; **shunt** (palpate along its course; press the reservoir only if trained).
5. **Face**: symmetry at rest and when crying or smiling (cranial nerve VII, step E).
6. **Neck stiffness**: with the child supine, gently flex the neck (or ask an older child to "kiss your knees" or look at a toy on the belly); resistance or pain = positive. Not in suspected neck trauma.
7. **Kernig's sign**: hip and knee flexed to 90°, then extend the knee: pain and resistance in the hamstrings or back = positive.
8. **Brudzinski's sign**: passive neck flexion causes involuntary flexion of the hips and knees.
9. **Infants**: bulging fontanelle, irritability when handled ("paradoxical irritability": settles when left still, cries when picked up), high-pitched cry, opisthotonus.

E. Cranial nerves (adapted to age)

Nerve	Infant / toddler	Older child
I	Not tested	Smell (rarely needed)
II	Fixes and follows a face or bright toy (by 3 months); **red reflex**; pupils	Acuity (picture or letter chart), fields by confrontation with a toy, pupils, **fundoscopy**
III, IV, VI	Follows a toy in all directions; corneal light reflex and **cover test** for squint; nystagmus; ptosis	Eye movements in an "H" pattern, diplopia, nystagmus; **sixth-nerve palsy** = false localising sign of raised ICP
V	Corneal reflex (only if comatose), rooting, suck	Facial sensation, jaw clench
VII	Facial symmetry when crying or smiling (UMN spares the forehead; LMN or Bell's affects the whole half)	Raise eyebrows, close eyes tight, show teeth, puff cheeks
VIII	Startles or turns to sound (distraction test from behind, out of sight)	Whisper test, finger rub; Rinne and Weber in older children
IX, X	Suck, swallow, cry, gag (if indicated), drooling, nasal regurgitation	Palate elevation ("say aah"), voice, swallow
XI	Head control	Shrug shoulders, turn head against resistance
XII	Tongue at rest (fasciculation in SMA), sucking	Protrude tongue (deviates towards the weak side), move side to side

Fundoscopy is distressing: do it near the end (step J), after dilating the pupils if possible in a comatose child.

> **UGANDA CONTEXT**
> In a comatose child with malaria, look at the **fundi** after dilating the pupils if possible: **malarial retinopathy** (patchy retinal whitening, vessel colour changes, white-centred haemorrhages, papilloedema) strongly supports **cerebral malaria** as the cause of coma; its absence should make you think harder about meningitis, encephalitis, hypoglycaemia or poisoning. Retinal haemorrhages without malaria in an infant with an unexplained head injury raise concern for **abusive head trauma**.

F. Motor system (inspection → tone → power → reflexes → coordination → sensation)

1. **Inspection**: posture (frog-leg hypotonia, scissoring, fisting with adducted thumb, hemiplegic posture), **muscle bulk** (wasting; calf pseudohypertrophy in Duchenne), **fasciculation** (tongue, limbs: SMA), involuntary movements, asymmetry, contractures.
2. **Tone**:
   - **Infants**: **pull-to-sit** (head lag beyond 4 months is abnormal), **ventral suspension** (hold prone over your hand: a floppy baby drapes like an inverted "U"), **vertical suspension** (hold under the arms: a hypotonic baby slips through your hands; a spastic baby **scissors** the legs and points the toes), "scarf sign", passive limb movement.
   - **Older children**: relaxed limbs, roll the leg, lift the knee quickly (catch), flex and extend elbow and knee at different speeds (spasticity is velocity-dependent; rigidity is not). Grade with the **Modified Ashworth Scale** (section 3).
   - **Ankle clonus**: brisk dorsiflexion and hold; > 3–5 beats sustained = abnormal.
3. **Power**:
   - **Infants and toddlers**: functional: kicks against gravity, reaches, bears weight, sits, stands, climbs; **Gowers sign** (rising from the floor by "climbing up" the thighs: proximal weakness, for example Duchenne).

- **Older children**: **MRC scale** for each myotome (section 3); pronator drift.

4. **Reflexes**: child relaxed, compare sides, use reinforcement (clench teeth, Jendrassik). Biceps, supinator, triceps, knee, ankle; grade 0–4+. **Plantar response**: stroke the lateral sole with a blunt object; upgoing is normal until about 12–18 months.
5. **Primitive and protective reflexes** (infants): Moro, rooting, palmar and plantar grasp, ATNR, stepping, **parachute** (from 6–9 months). Note persistence and asymmetry.
6. **Coordination**: infants: reaching accuracy, tremor; older: finger–nose, heel–shin, rapid alternating movements (dysdiadochokinesia), **tandem (heel-to-toe) walking**, Romberg (sensory ataxia).
7. **Sensation** (older children, or if a spinal level is suspected): light touch, pain (blunt or sharp with care), vibration, proprioception; look for a **sensory level** (cord lesion) and saddle anaesthesia.

G. Gait

Watch the child walk in shorts and bare feet, then on tiptoes, on heels, run, hop, and (older) heel-to-toe. Interpretation is in section 6.

H. Spine and back

Inspect from behind: midline skin markers (**sacral dimple above the natal cleft, hair tuft, lipoma, haemangioma, sinus**: occult spinal dysraphism), **scoliosis** (shoulder and hip asymmetry; **Adam's forward-bend test** shows a rib hump), **kyphosis or gibbus** (**TB spine, Pott's disease**), tenderness, range of movement.

I. Musculoskeletal: pGALS screen (school-age child), then look–feel–move

Three screening questions: (1) Do you (or your child) have any pain or stiffness in your joints, muscles or back? (2) Can you (or your child) dress yourself completely without any difficulty? (3) Can you (or your child) walk up and down stairs without any difficulty?

Manoeuvre	What it tests
Observe standing (front, back, side)	Posture, alignment (genu varum or valgum), swelling, muscle wasting, leg-length discrepancy, scoliosis
Walk; walk on tiptoes; walk on heels	Gait, ankles, subtalar and foot joints, foot drop
"Hold your hands out straight in front of you"	Forward flexion of shoulders, elbow extension, wrist extension, finger extension
"Turn your hands over and make a fist"	Supination, small joints of fingers
"Pinch your index finger and thumb together"	Fine motor, index finger and thumb joints
"Touch the tips of your fingers with your thumb"	Manual dexterity, small finger joints
Squeeze the metacarpophalangeal joints	Tenderness (inflammatory arthritis)
"Put your hands together palm to palm, then back to back"	Wrist and finger extension and flexion
"Reach up and touch the sky; look at the ceiling"	Shoulders, elbows, wrists, neck extension
"Put your hands behind your neck"	Shoulder abduction and external rotation
"Try to touch your shoulder with your ear"	Cervical lateral flexion
"Open your mouth and put 3 of your own fingers in"	Temporomandibular joints (often affected in JIA)
Feel for a knee effusion (patellar tap, bulge test); active knee flexion and extension	Knee swelling and range
Passive hip flexion and internal rotation (lying)	Hip (loss of internal rotation is an early sign of hip disease)

Manoeuvre	What it tests
"Bend forward and touch your toes"	Thoracolumbar flexion, scoliosis

For any abnormal joint: **Look** (swelling, redness, deformity, wasting, scars), **Feel** (warmth, tenderness, effusion, bony tenderness at the metaphyses in osteomyelitis), **Move** (active, then passive range; pain; crepitus), then examine the joint above and below and do a **neurovascular** check. In infants examine the **hips** (Ortolani and Barlow up to about 3 months; then limited abduction, leg-length difference, asymmetric thigh creases, Galeazzi sign). Examine the painful joint **last** and gently.

J. Completing the examination

1. **Fundoscopy** (if not already done) and any remaining distressing parts.
2. "To complete my examination I would like to: **plot the head circumference and growth**, examine the other systems (abdomen for organomegaly, heart for murmurs, skin), perform **fundoscopy**, check the **glucose**, assess **development**, and review the history and investigations."
3. Cover the child, **thank** the child and parent, **wash hands**.
4. **Explain findings** to the parents in simple words and **summarise** with a localisation (section 9).

Focusing the examination by presentation

Presentation	Must-do elements
Coma or convulsions	ABCD, Blantyre or GCS, pupils, posture, fontanelle, meningism, **fundi**, focal signs, glucose
Weakness	Pattern (UMN versus LMN), reflexes, sensation and level, bladder, cranial nerves (bulbar), breathing
Developmental delay	Observation, tone, reflexes, primitive reflexes, HC, dysmorphism, vision and hearing
Limp or joint pain	Gait, pGALS, look–feel–move, hips, spine, neurovascular

6. Findings and Interpretation

UMN, LMN or muscle?

Feature	UMN	LMN	Muscle
Tone	**Increased** (spastic catch)	Reduced	Normal or reduced
Reflexes	**Brisk**, clonus > 3–5 beats	Reduced or **absent**	Normal or reduced
Plantar	**Upgoing** (after 12–18 months)	Flexor or absent	Flexor
Bulk	Little wasting	**Wasting, fasciculation**	Proximal wasting; calf pseudohypertrophy (Duchenne)
Weakness pattern	Pyramidal	Distal or segmental	**Proximal**; **Gowers sign**

Key signs and what they mean

Finding	What it means
AVPU P or U; BCS ≤ 2; GCS ≤ 8	Coma: airway at risk
Bulging fontanelle (upright, calm child)	Raised ICP, meningitis
Sunken fontanelle	Dehydration

Finding	What it means
Unequal or poorly reactive pupils, abnormal posturing, focal paralysis, irregular breathing, bradycardia with hypertension, papilloedema, deteriorating coma score	**Raised ICP (WHO): no LP**
Sixth-nerve palsy	False localising sign of raised ICP
Facial weakness sparing the forehead	UMN VII lesion; whole half of the face = LMN (Bell's palsy)
Tongue fasciculation, frog-leg posture, alert face, absent reflexes	SMA (peripheral hypotonia)
Persistent primitive reflexes	CP
Asymmetric Moro	Brachial plexus injury, clavicle fracture, hemiplegia
Head lag after 4 months	Hypotonia or weakness
Scissoring on vertical suspension	Spasticity
Sensory level, saddle anaesthesia, urinary retention	Spinal cord lesion
Midline sacral dimple, hair tuft, lipoma, sinus	Occult spinal dysraphism
Gibbus	Pott's disease (TB spine)
Loss of hip internal rotation	Early hip disease
Bony tenderness at the metaphysis	Osteomyelitis
Hip held flexed, abducted and externally rotated	Septic arthritis of the hip (maximises capsule volume)
Wide wrists, rachitic rosary, progressive bowing	Rickets

Gait

Gait	Features	Causes (Uganda examples)
Antalgic	Short stance on the painful leg	Septic arthritis, osteomyelitis, pyomyositis, fracture, sickle crisis
Trendelenburg / waddling	Pelvis drops on the opposite side when standing on the affected leg; bilateral = waddling	DDH, Perthes disease, slipped capital femoral epiphysis, proximal myopathy (Duchenne), rickets
Hemiplegic	Arm flexed, leg extended and circumducted, toe-walking on one side	Hemiplegic CP, **sickle cell stroke**, post-meningitis
Diplegic / scissoring	Both legs stiff, adducted, toe-walking	Spastic diplegia (preterm PVL), HIV encephalopathy
Ataxic	Broad-based, unsteady, cannot tandem walk	Cerebellar lesion, post-malaria cerebellar ataxia, drug toxicity (phenytoin), posterior fossa tumour
High-stepping (foot drop)	Lifts the knee high, foot slaps	**Sciatic or common peroneal nerve injury from IM injection** in the buttock, Guillain–Barré syndrome, polio sequelae
Toe walking	Persistent tiptoe	CP, Duchenne, tight Achilles tendons, habit (idiopathic), autism

Differential diagnosis by finding

Finding	Differentials to consider
Coma with fever	Cerebral malaria, bacterial meningitis, TB meningitis, viral encephalitis, sepsis, post-ictal state, hypoglycaemia
Coma without fever	Poisoning (organophosphates, alcohol, benzodiazepines), hypoglycaemia, head injury (including abuse), DKA, uraemia, intracranial bleed, status epilepticus
Acute flaccid weakness	Guillain–Barré syndrome, poliomyelitis or non-polio enterovirus, transverse myelitis, injection neuritis, hypokalaemia, botulism, tick paralysis, spinal cord compression (early)
Spastic weakness	CP, HIV encephalopathy, stroke (sickle cell), spinal cord lesion (TB spine, tumour), hereditary spastic paraplegia, konzo (cassava-related, regional)
Floppy infant	Central (HIE, Down syndrome, sepsis, hypothyroidism, metabolic) versus peripheral (SMA, congenital myopathy, myasthenia)
Ataxia	Post-infectious cerebellitis (varicella), drug toxicity (phenytoin, carbamazepine), posterior fossa tumour, cerebral malaria sequelae
Painful swollen joint	Septic arthritis, osteomyelitis (adjacent), sickle crisis, rheumatic fever (migratory), JIA, haemophilia, trauma, leukaemia
Limp	Septic arthritis, transient synovitis, osteomyelitis, pyomyositis, Perthes, SCFE, DDH, fracture, injection injury, leukaemia, Burkitt lymphoma

Bedside tests

Test	What it tells you
Glucose	Hypoglycaemia (coma, seizures): the commonest treatable cause with malaria
Malaria RDT or smear	Cerebral malaria, malaria with seizures
SpO_2, RR, peak cough or vital capacity (older child), single-breath count	Respiratory muscle weakness in Guillain–Barré syndrome
Hb (HemoCue), sickle test	Severe anaemia; sickle cell disease in stroke or dactylitis
Urine dipstick, bladder scan or palpation	Retention in cord compression
Fundoscopy	Papilloedema, malarial retinopathy, retinal haemorrhages
Head circumference plotted	Micro- or macrocephaly, crossing centiles
Joint aspiration (by a trained clinician)	Pus in septic arthritis (Gram stain, culture)

Laboratory and imaging

Test	When	Expected or abnormal finding	How it changes management
Lumbar puncture (CSF)	Meningism, coma without contraindication, suspected GBS (after a week)	Bacterial: neutrophils, low glucose, high protein; TB: lymphocytes, very high protein, low glucose; GBS: high protein with normal cells	Antibiotics, TB treatment, IVIG. HC IV (if trained) and hospitals; culture at RRH and NRH
FBC, CRP or ESR, blood culture	Fever, joint or bone infection	Leucocytosis, raised CRP or ESR	Antibiotics; monitoring response
Electrolytes, calcium, magnesium	Seizures, weakness, tetany	Hyponatraemia, hypocalcaemia (rickets), hypokalaemia (periodic paralysis)	Correction

Test	When	Expected or abnormal finding	How it changes management
CK	Proximal weakness, Gowers sign, calf hypertrophy	Very high in Duchenne; high in myositis	Diagnosis, genetic counselling
Stool samples × 2 (24 h apart, within 14 days)	**Acute flaccid paralysis**	Poliovirus or non-polio enterovirus	Public health response; sent via the district surveillance officer to the national polio laboratory (UVRI); eradication surveillance
Joint or bone X-ray	Limp, swelling, deformity, rickets	Effusion, periosteal reaction (osteomyelitis after about 10–14 days), metaphyseal cupping and fraying (rickets), fracture	Surgical drainage, rickets treatment. All hospitals
Ultrasound of a joint (hip)	Painful hip	Effusion	Guided aspiration
Spine X-ray	Gibbus, back pain, scoliosis	Vertebral collapse, disc space loss (TB), Cobb angle	TB treatment, orthopaedic or neurosurgical referral
CT or MRI brain or spine	Focal signs, raised ICP, suspected abscess, stroke, cord compression	Infarct, abscess, tumour, hydrocephalus, cord compression	Neurosurgery, stroke management. CT at RRH, NRH, Mengo; MRI in Kampala
EEG	Seizures of unclear type, encephalopathy, nodding syndrome	Epileptiform activity	Antiepileptic choice. Mulago, CoRSU, private
Nerve conduction studies	GBS, neuropathy	Demyelination or axonal loss	Diagnosis. Very limited (Mulago, private)

7. Clinical Acumen

What experienced clinicians look for first

- **Is the child sick?** (ETAT, coma score, glucose).
- **Is there meningism or raised ICP?** (determines LP and urgency).
- **Is there a focal sign?** (pupils, eye movements, facial weakness, hemiparesis) → structural lesion, stroke, abscess, TB meningitis, focal seizure with Todd's paresis.
- **Is it UMN, LMN or muscle?** (tone, reflexes, plantar, wasting, fasciculation, Gowers).
- **Is there a spinal level?** (sensory level, bladder, saddle anaesthesia, back pain, gibbus) → cord compression is an emergency.
- **Is the head size normal and the fontanelle normal?**
- **Is there a hot joint?** → septic arthritis until proven otherwise.

Red flags

RED FLAG

Act now if you find:

- **AVPU P or U, Blantyre ≤ 2 or GCS ≤ 8**: airway, glucose, treat fits, ETAT coma pathway.
- **Signs of raised ICP**: unequal or fixed pupils, posturing, irregular breathing, bradycardia with hypertension, papilloedema → no LP, head-up 30°, senior or ICU.
- **Meningism with fever or a petechial or purpuric rash** → antibiotics within the hour.
- **Acute flaccid weakness** (especially ascending, with respiratory or bulbar involvement) → AFP notification; monitor breathing (Guillain–Barré, polio).
- **Spinal cord signs** (sensory level, urinary retention, back pain, gibbus) → urgent imaging (TB spine, tumour, Burkitt lymphoma).
- **Hot, swollen, painful joint with fever**, or a child refusing to bear weight → septic arthritis or osteomyelitis.
- **Developmental regression** or a rapidly enlarging head.
- **Unexplained bruises, retinal haemorrhages, or fractures in a young child** → safeguarding.

Thinking like a clinician

- **Floppy infant**: alert and weak with absent reflexes and tongue fasciculation = **peripheral** (SMA); poor responsiveness with preserved reflexes and dysmorphism = **central** (HIE, Down syndrome, hypothyroidism, sepsis).
- **Ascending symmetrical weakness with areflexia** days after a diarrhoeal or respiratory illness = **Guillain–Barré syndrome**; **asymmetric flaccid paralysis with fever** in an under-immunised child = **polio** (or non-polio enterovirus): both are AFP and must be notified.
- **Unilateral foot drop after an injection in the buttock** = **sciatic nerve injection injury** (common in Uganda after IM quinine or other injections); UCG advises IM injections into the anterolateral **thigh**, never the buttock, in children.
- **Child with sickle cell disease and sudden hemiparesis** = **stroke** until proven otherwise; with **dactylitis** (painful swollen hands and feet) in infants, or **bone pain and fever** (osteomyelitis versus vaso-occlusive crisis; Salmonella).
- **Child with a limp and fever**: septic arthritis of the hip (holds the hip flexed, abducted and externally rotated, refuses to move) versus transient synovitis (well, can bear some weight) versus osteomyelitis versus pyomyositis (Uganda: very common, thigh or buttock swelling).
- **Bowed legs**: physiological genu varum is normal up to about **2 years** (and genu valgum at 3–5 years); persisting, progressive, asymmetric bowing or wide wrists and a rachitic rosary = **rickets** (or Blount disease).
- **Gibbus with back pain, weight loss and leg weakness** = **Pott's disease** (spinal TB): start TB treatment and refer to protect the cord.
- **Nodding syndrome** (northern Uganda, ages 5–15): repeated head-nodding atonic seizures triggered by food or cold, cognitive decline, stunting; associated with *Onchocerca volvulus*; covered in UCG 2023 (section 9.1.2) Verify *UCG 2023 section number*.
- **Cerebellar ataxia after cerebral malaria** is usually transient; persistent ataxia needs imaging.

When to stop and escalate

- Stop the full examination and treat if the child becomes unstable, convulses or deteriorates (falling coma score).
- Call a senior at once for raised ICP, coma, AFP with breathing or swallowing difficulty, cord signs, a suspected septic hip, or concerns about abuse.

Common student mistakes

- Starting with the reflex hammer in a sick child instead of ABCD and the conscious level.
- Reporting only the total coma score.
- Relying on neck stiffness in an infant.
- Calling an upgoing plantar in a 9-month-old abnormal.
- Testing neck stiffness after trauma.
- Doing the painful parts first and losing the child's cooperation.
- Forgetting head circumference, fundoscopy and glucose.

Worked example

EXAM FAVOURITE

Worked example: localising weakness. A 6-year-old has had 4 days of progressive weakness starting in the legs and now the arms, after a diarrhoeal illness 2 weeks ago. Findings: symmetrical weakness (legs MRC 2/5, arms 4/5), **absent reflexes**, flexor plantars, no sensory level, bladder normal, single-breath count 12. **Interpretation:** LMN pattern, ascending, areflexic, no cord level = **Guillain–Barré syndrome** (an AFP case: notify and send stool samples). A falling breath count signals impending respiratory failure: monitor 4-hourly, refer to HDU or ICU, specialist IVIG if available. Compare: a **sensory level at T10 with brisk reflexes and urinary retention** = cord compression (TB spine, tumour): urgent imaging.

8. Complications and Troubleshooting

Complications

Of the examination (rare)	Of missing findings
Distress and loss of the child's cooperation (do painful parts last)	Missed raised ICP → coning after LP
Bruising from excessive painful stimuli	Missed meningitis → death or deafness
Neck movement in an unrecognised cervical spine injury	Missed AFP → missed polio case and outbreak
Worsening pain from forced movement of an inflamed joint	Missed septic hip → avascular necrosis, joint destruction
	Missed spinal cord compression → permanent paraplegia

Troubleshooting

- **Crying infant**: start with observation and play; ask the parent to help; come back when settled. A crying baby gives you nothing.
- **Uncooperative toddler**: make it a game (soldier walk, tiptoes, hop, "touch my finger").
- **Reflexes hard to elicit**: relax the child, use reinforcement (clench teeth, Jendrassik).
- **Child cannot be fully examined because unstable**: do the focused examination (section 2) and document what remains.

Acting on what you find

The examination itself has no complications if done gently, but **what you find must lead to action**.

Finding	Immediate action
Coma (AVPU P or U, BCS ≤ 2, GCS ≤ 8)	Airway (recovery position, suction, oxygen; airway adjunct if needed), **glucose** (10% dextrose 5 mL/kg if < 3.0 mmol/L), treat seizures, **IV artesunate** (3 mg/kg if < 20 kg; 2.4 mg/kg if ≥ 20 kg) if malaria positive, **ceftriaxone 100 mg/kg/day** if meningitis not excluded, maintenance fluids (avoid overload), 2-hourly neuro-observations (BCS, pupils), turn 2-hourly, NG feeds, eye care
Signs of raised ICP	**Do not do an LP**; head-up 30°, head midline; avoid hypotonic fluids and fluid overload; treat fever and seizures; oxygen; senior and anaesthetic or ICU review; CT when stable; specialist mannitol or hypertonic saline only on senior advice (WHO advises against routine mannitol or steroids in cerebral malaria)
Meningism	LP if no contraindication; start **ceftriaxone 100 mg/kg/day** (max 4 g) without delay (UCG 2023)
Acute flaccid paralysis (< 15 years)	Notify the district surveillance focal person; **2 stool samples 24 h apart within 14 days** (kept cold); **no IM injections** (UCG); monitor breathing and swallowing; physiotherapy and positioning; GBS → HDU or ICU monitoring, IVIG where available
Spinal cord compression	Lie flat, catheterise if retention, urgent imaging and neurosurgical or orthopaedic referral; start TB treatment if Pott's disease is likely; dexamethasone for tumour only on specialist advice
Focal deficit in sickle cell disease	Oxygen, hydration, urgent **exchange (or simple) transfusion** per protocol, CT, then a long-term transfusion or hydroxyurea programme
Septic arthritis or osteomyelitis	Analgesia, splint, blood culture, **joint aspiration** or surgical drainage (hip = surgical emergency), **cloxacillin 50 mg/kg IV 6-hourly** (UCG 2023), adding **ceftriaxone** in children with a negative Gram stain or sickle cell disease (Salmonella); IV for 2 weeks then oral to complete 4 weeks (longer for osteomyelitis)
Rickets	Vitamin D and calcium, diet and sunlight advice, follow-up X-ray and ALP
Developmental delay or CP signs	HIV test, hearing and vision, MDT referral (see Cerebral Palsy and Growth & Development topics)
Sciatic nerve injection injury	Examine power and sensation, physiotherapy and an AFO, consider nerve studies, report it
Unexplained injuries or retinal haemorrhages	Safeguarding: admit, document carefully, skeletal survey, inform the senior and the social worker or probation officer

Drugs you may start after the examination

Drug	Dose	Route	Frequency	Duration / max	Notes
10% dextrose	**5 mL/kg** (neonate 2 mL/kg)	IV	Bolus; recheck in 30 min		For glucose < 3.0 mmol/L (UCG; WHO < 2.5, < 3.0 in SAM)
Diazepam	**0.5 mg/kg** (rectal) or **0.2–0.3 mg/kg** (IV)	PR or IV	Once; repeat once after 10 min	**Max 10 mg/dose**; 2 doses total	For a seizure ≥ 5 min
Artesunate	**3 mg/kg** (< 20 kg); **2.4 mg/kg** (≥ 20 kg)	IV (or IM)	0, 12, 24 h, then daily	Until oral ACT tolerated	Severe or cerebral malaria
Ceftriaxone	**100 mg/kg/day**	IV or IM	Once daily or divided 12-hourly	**Max 4 g/day**; 10–14 days for meningitis	Pre-referral IM dose at HC level

Drug	Dose	Route	Frequency	Duration / max	Notes
Cloxacillin	**50 mg/kg/dose**	IV	**6-hourly**	2 weeks IV, then oral to 4 weeks total (adult max 2 g/dose)	Septic arthritis, osteomyelitis, pyomyositis (UCG 2023)
Paracetamol	**15 mg/kg**	Oral or rectal	6-hourly	Max 60 mg/kg/day	Pain, fever
Ibuprofen	**5–10 mg/kg**	Oral	6–8-hourly	Max 30–40 mg/kg/day	Musculoskeletal pain (avoid if dehydrated, renal impairment, bleeding)

9. Documentation and Presenting Findings

Document these positives	And these negatives
AVPU and Blantyre or GCS with components (for example "BCS 2 = M1 V1 E0") and the time	"Pupils equal and reactive, no papilloedema"
Tone, power, reflexes per limb; plantar responses	"No neck stiffness, fontanelle soft"
Head circumference with centile	"No sensory level; bladder not palpable"
Joint findings (swelling, warmth, range)	"No spinal deformity or midline skin markers"

Model summary of findings

> "On examination, a 3-year-old boy, Blantyre 2 (motor 1, verbal 1, eyes 0) one hour after his last seizure, with equal reactive pupils, no papilloedema but bilateral malarial retinopathy (retinal whitening and haemorrhages), no neck stiffness, normal tone with symmetrical brisk reflexes and flexor plantars, no focal deficit and no evidence of raised ICP; glucose 5.2 mmol/L; malaria smear positive. Summary: unrousable coma with malarial retinopathy, consistent with cerebral malaria; meningitis not excluded."

Model case presentation (ward round)

> "Akello is a 3-year-old girl from Masaka weighing 12 kg (WAZ −1.0, HAZ −0.8, WHZ −0.9, MUAC 14.5 cm), fully immunised per the UNEPI card, HIV-negative, on a family diet, admitted 6 hours ago with 2 days of fever and 3 generalised seizures. On neurological examination 1 hour after the last seizure, she is responding to pain only; Blantyre coma scale 2 (withdraws from pain, M1; moans, V1; not following, E0). Pupils are 3 mm, equal and reactive; fundoscopy shows patchy retinal whitening and white-centred haemorrhages. There is no neck stiffness, Kernig's sign is negative, and there are no signs of raised ICP. Tone is normal, reflexes are symmetrical and brisk, plantars are flexor, with no focal deficit. Glucose 4.6 mmol/L; malaria smear 4+ (180,000/µL); Hb 7.2 g/dL. **Impression: cerebral malaria (BCS 2 with malarial retinopathy) with repeated convulsions; bacterial meningitis not excluded.** Plan: IV artesunate 36 mg (3 mg/kg) at 0, 12 and 24 h, LP when stable (no contraindication now) or ceftriaxone 1.2 g/day empirically, seizure management, 2-hourly BCS and glucose, maintenance fluids, NG feeds after 24 h if still unconscious."

> **NOTE**
> In the model presentation above the motor score is 1 (withdraws), so BCS = M1 + V1 + E0 = 2. Always state the **components**, not just the total; it shows you know the scale and allows others to track change.

Structure of the summary: level (cerebral, cerebellar, spinal, nerve, muscle), side, coma score, likely cause, and the next investigation.

10. OSCE Checklist

Step	Done well when	Critical?
Introduction and consent	Washes hands, introduces self, confirms the child's name and age, gains consent from parent and assent from the child	Yes
Privacy and comfort	Ensures privacy and appropriate exposure, parent present, child comfortable	No
General inspection	ABCD, conscious level, posture, movements, dysmorphism, neurocutaneous stigmata, aids	Yes
Conscious level	States or assesses AVPU and the Blantyre coma scale (or GCS) correctly **with components**	Yes
Head circumference	Measures correctly and states it should be plotted	Yes
Fontanelle and meningism	Examines fontanelle and sutures (infant); tests neck stiffness and Kernig or Brudzinski (older child)	No
Cranial nerves	Age-appropriate (fix and follow, eye movements, face, hearing, swallowing, tongue); mentions fundoscopy	No
Motor inspection	Muscle bulk, fasciculation, involuntary movements	No
Tone	Pull-to-sit, ventral and vertical suspension (infants); passive movements and clonus (older children)	No
Power	Functional (infant) or MRC grading (older child); looks for Gowers sign	No
Reflexes	Biceps, triceps, supinator, knee, ankle and plantars elicited correctly, comparing sides	Yes
Primitive reflexes or coordination	Tested as appropriate for age	No
Gait	Normal, tiptoe, heel; running or hopping if age allows	No
Spine	Midline markers, scoliosis with forward bend, gibbus	No
Joints	pGALS screen or look–feel–move; hips in infants	No
Approach	Gentle, plays with the child, leaves painful parts until last	Yes
Closure	Thanks the child and parent, covers the child	No
Presentation	Structured summary (UMN, LMN or muscle; localisation; coma score) with a differential and next steps	Yes

11. Key Points

High-yield points

- In a sick child: **ABCD, AVPU and glucose** before the reflex hammer.
- **Blantyre 2-2-1** (motor, verbal, eyes; max 5); ≤ 2 = coma; always give components.
- **Observe and play** first; painful parts last.
- **Measure and plot head circumference** in every child.
- Meningism is unreliable under 12–18 months: look at the fontanelle and behaviour.
- **Raised ICP = no LP**; treat empirically.
- **UMN everything up, LMN everything down**, muscle = proximal weakness and Gowers.
- **AFP in a child < 15 years**: notify, 2 stools 24 h apart within 14 days, no IM injections.
- **Hot joint + fever = septic arthritis** until proven otherwise: aspirate and give cloxacillin.

- **Fundoscopy in coma**: malarial retinopathy supports cerebral malaria.

Numbers & doses

Item	Value
Blantyre coma scale	Max **5** (motor 2, verbal 2, eyes 1); **≤ 2 = coma** (cerebral malaria)
GCS	3–15; **≤ 8** = protect airway
AVPU	**P or U** = coma (ETAT emergency)
Head lag	Abnormal after **4 months**
Primitive reflexes	Gone by **4–6 months**; parachute from **6–9 months**
Upgoing plantar	Normal until **12–18 months**
Clonus	**> 3–5 beats** sustained = abnormal (beyond the newborn period)
Reflex roots	Biceps **C5–6**, triceps **C7**, knee **L3–4**, ankle **S1**
MRC power	**0–5** (3 = against gravity)
Physiological bowing	Genu varum to **~2 years**; genu valgum **3–5 years**
Neurofibromatosis type 1	**≥ 6** café-au-lait macules > 5 mm (prepubertal) or > 15 mm (postpubertal)
AFP surveillance	Child **< 15 years**; **2 stools 24 h apart within 14 days**; no IM injections
Hypoglycaemia	**< 3.0 mmol/L** (UCG) → **10% dextrose 5 mL/kg**
Ceftriaxone	**100 mg/kg/day** (max 4 g)
Cloxacillin	**50 mg/kg IV 6-hourly**, 2 weeks IV then oral to 4 weeks (septic arthritis)
IV artesunate	**3 mg/kg** (< 20 kg) / **2.4 mg/kg** (≥ 20 kg) at 0, 12, 24 h then daily
Diazepam	**0.5 mg/kg PR** / **0.2–0.3 mg/kg IV**, max 10 mg

Exam traps

- **"Neck stiffness is absent, so this 8-month-old does not have meningitis."** Wrong: meningism is unreliable under 12–18 months; look at the fontanelle and behaviour.
- **"Upgoing plantar in a 9-month-old means UMN disease."** Wrong: normal until about 12–18 months.
- **"Do an LP to diagnose meningitis in a child with unequal pupils."** Wrong: raised ICP; treat empirically, LP later.
- **"Test neck stiffness in a child after a fall from a tree."** Wrong: immobilise the cervical spine first.
- **"Report only total coma scores."** Wrong: report components.
- **"Give an IM injection to a child with acute flaccid paralysis."** Wrong: IM injections can worsen paralysis (provocation polio); UCG says avoid.
- **"Bowed legs at 18 months = rickets."** Not necessarily: physiological genu varum is normal until about 2 years; look for other rickets signs.
- **"Septic hip can wait for an X-ray to show changes."** Wrong: X-rays are normal early; ultrasound and aspiration urgently.

Questions seniors ask

Viva questions

1. *Describe the Blantyre coma scale.* Motor 0–2 (localises, withdraws, none), verbal 0–2 (appropriate cry, moan, none), eyes 0–1 (watches or follows, or not); total 5; ≤ 2 = coma.

2. *How do you distinguish UMN from LMN weakness?* UMN: increased tone, brisk reflexes, clonus, upgoing plantars, little wasting. LMN: decreased tone, reduced or absent reflexes, wasting, fasciculation.
3. *When is an upgoing plantar normal?* In infants until about 12–18 months.
4. *What is Gowers sign?* Using the hands to "climb up" the legs to stand from the floor: proximal (pelvic girdle) weakness, classically Duchenne.
5. *Signs of raised ICP in a child?* Bulging fontanelle, unequal pupils, posturing, focal deficit, irregular breathing, bradycardia with hypertension, papilloedema, sixth-nerve palsy, falling coma score.
6. *What do you do with a case of AFP?* Notify, collect 2 stool samples 24 h apart within 14 days, avoid IM injections, monitor breathing.
7. *Name the pGALS screening questions.* Pain or stiffness in joints, muscles or back; difficulty dressing; difficulty with stairs.
8. *Why is the hip held flexed, abducted and externally rotated in septic arthritis?* This position maximises the joint capsule volume and reduces pain from the effusion.

On the ward round

- **How would you assess the conscious level of a 2-year-old with cerebral malaria?** AVPU first, then the Blantyre coma scale with components (motor 0–2 with pencil pressure on the sternum and nail bed, verbal 0–2, eyes 0–1); repeat 2-hourly; ≤ 2 = coma.
- **This child has brisk reflexes and upgoing plantars on one side. Where is the lesion?** Upper motor neurone, contralateral cerebral hemisphere or brainstem (or ipsilateral spinal cord if there is a sensory level); look for a cause such as stroke (sickle cell), abscess, TB meningitis or a post-ictal Todd's paresis.
- **How do you examine tone in a 6-month-old?** Pull-to-sit (head control), ventral suspension (posture over the hand), vertical suspension (slipping through or scissoring), passive limb movements, and check for persistent primitive reflexes.
- **What would make you think a floppy infant has a neuromuscular rather than central cause?** Alert face with good social interaction, marked weakness, absent reflexes, tongue fasciculation, frog-leg posture (SMA); central causes have reduced alertness, dysmorphism or seizures with preserved or brisk reflexes.
- **A 7-year-old walks with a high-stepping gait on the right after a "quinine injection" last month. What happened and what do you do?** Sciatic (or common peroneal) nerve injection injury from an IM buttock injection; examine power and sensation, refer for physiotherapy and an AFO, consider nerve studies, and report it; prevent by giving IM injections in the anterolateral thigh.
- **What would you do for a 10-year-old with a hot, swollen knee and fever?** Analgesia, splint, FBC, CRP and blood culture, urgent aspiration (Gram stain, culture), IV cloxacillin 50 mg/kg 6-hourly (plus ceftriaxone if sickle cell disease or Gram-negative), orthopaedic review for washout.
- **Why examine the fundi in a comatose child in Uganda?** Malarial retinopathy supports cerebral malaria; papilloedema suggests raised ICP (no LP); retinal haemorrhages without malaria in an infant raise concern for abusive head trauma.

Mnemonics

- **Reflex roots "1-2, buckle my shoe (S1–2 ankle); 3-4, kick the door (L3–4 knee); 5-6, pick up sticks (C5–6 biceps and supinator); 7-8, lay them straight (C7–8 triceps)."**
- **Blantyre "2-2-1"**: Motor 2, Verbal 2, Eyes 1 = maximum 5.
- **UMN = "everything UP"**: tone up, reflexes up, plantars up. **LMN = "everything DOWN"**: tone down, reflexes down, muscle bulk down (plus fasciculations).
- **Raised ICP in infants "FIBS"**: **F**ontanelle bulging, **I**rritability or high-pitched cry, **B**ig head (crossing centiles), **S**unsetting eyes and split sutures.
- **Septic hip, Kocher criteria "FEW-W"**: **F**ever > 38.5 °C, **E**SR > 40 mm/h, **W**hite cells > 12 × 10^9/L, **W**on't bear weight. The more present, the more likely septic arthritis (all 4: very likely).

- **AFP "2-24-14"**: **2** stool samples, **24** h apart, within **14** days.
- **WIPER**: Wash, Introduce, Privacy, Expose, Reposition.

OSCE tips

- **Narrate**: "I am now assessing tone by…" helps the examiner follow you.
- In an infant, **start with observation and play**; a crying baby gives you nothing. Ask the parent to help.
- Always mention **head circumference, fundoscopy and glucose** even if you are not asked to perform them.
- Finish with a **localising summary**: level (cerebral, cerebellar, spinal, nerve, muscle), side, likely cause, and the next investigation.

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SELF-TEST

Flashcards

28 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What is the first step in the neurological examination of a sick child?	**ETAT ABCD and conscious level** (AVPU), plus **glucose**; stabilise before the detailed examination.
2	AVPU: which levels mean coma?	**P** (responds only to pain) or **U** (unresponsive) = coma, an ETAT emergency sign.
3	Components of the Blantyre coma scale?	**Motor** 2 localises / 1 withdraws / 0 none; **verbal** 2 appropriate cry / 1 moan or abnormal cry / 0 none; **eyes** 1 watches or follows / 0 fails. Total **0–5**.
4	Blantyre score defining unrousable coma (cerebral malaria)?	**≤2** (WHO); UCG 2023 defines deep coma as **unable to localise a painful stimulus**.
5	How do you apply a painful stimulus for the Blantyre motor score?	**Localise**: firm pressure with the blunt end of a pencil on the **sternum or supraorbital ridge**. **Withdraw**: pressure on a **nail bed** with a horizontal pencil.
6	GCS threshold for airway protection?	**GCS ≤8**.
7	Why are Kernig and Brudzinski signs unreliable in infants, and what do you look for instead?	Meningism is often absent under **12–18 months**; look for a **bulging fontanelle**, irritability (paradoxical), high-pitched cry, poor feeding, drowsiness.
8	WHO signs of raised ICP that contraindicate LP?	**Unequal pupils**, **rigid posture/posturing**, **paralysis of limbs or trunk**, **irregular breathing**; also bradycardia with hypertension, papilloedema, falling coma score.
9	UMN vs LMN signs?	**UMN**: increased tone, brisk reflexes, clonus, upgoing plantar, little wasting. **LMN**: hypotonia, reduced/absent reflexes, wasting, fasciculation.
10	Root values of the deep tendon reflexes?	Biceps **C5–6**, supinator **C5–6**, triceps **C7**, knee **L3–4**, ankle **S1**.
11	Reflex grading scale?	**0** absent, **1+** reduced, **2+** normal, **3+** brisk, **4+** brisk with clonus.
12	MRC power grades?	**0** none, **1** flicker, **2** moves with gravity eliminated, **3** against gravity, **4** against some resistance, **5** normal.
13	Until what age is an upgoing (extensor) plantar normal?	Until about **12–18 months** (once walking is established it is abnormal).
14	Three tests of tone in an infant?	**Pull-to-sit** (head lag abnormal after 4 months), **ventral suspension** (floppy = inverted U), **vertical suspension** (slips through = hypotonia; scissoring = spasticity).
15	What is Gowers sign and what does it indicate?	Child 'climbs up' his thighs with his hands to stand from the floor: **proximal (pelvic girdle) weakness**, classically **Duchenne muscular dystrophy**.
16	Clinical clues to a peripheral (neuromuscular) cause of floppiness?	**Alert** face and social, marked **weakness**, **absent reflexes**, **tongue fasciculation**, frog-leg posture (e.g. spinal muscular atrophy).
17	Neurocutaneous stigmata to look for?	**≥6 café-au-lait macules** (>5 mm pre-pubertal) and axillary freckling (NF1); **ash-leaf macules**, facial angiofibromas, shagreen patch (tuberous sclerosis); facial **port-wine stain** (Sturge–Weber).
18	What is malarial retinopathy and why look for it?	Patchy **retinal whitening**, vessel colour changes, **white-centred haemorrhages**, ± papilloedema; supports **cerebral malaria** as the cause of coma (absence → think of other causes).

#	Question	Answer
19	Action for acute flaccid paralysis in a child under 15 (UCG 2023)?	Treat as possible **polio**: notify the district surveillance focal person, **2 stool samples 24 h apart within 14 days** (cold chain), **no IM injections**, monitor breathing.
20	Classic features of Guillain–Barré syndrome?	**Ascending, symmetrical flaccid weakness** with **areflexia**, days–weeks after diarrhoea or respiratory infection; risk of **respiratory and bulbar failure**; CSF high protein, normal cells.
21	Cause of foot drop after an injection in the buttock?	**Sciatic (or common peroneal) nerve injection injury**; prevent by giving IM injections in the **anterolateral thigh** in children.
22	Three pGALS screening questions?	Any **pain or stiffness** in joints, muscles or back? Any difficulty **dressing**? Any difficulty with **stairs**?
23	Position of a septic hip and why?	**Flexed, abducted, externally rotated**: maximises capsular volume and reduces pain from the effusion.
24	Kocher criteria for septic arthritis of the hip?	**Fever >38.5°C**, **non-weight-bearing**, **ESR >40 mm/h**, **WCC >12 × 10^9/L**; probability rises with each criterion.
25	Empirical antibiotic for septic arthritis in a child (UCG 2023)?	**Cloxacillin 50 mg/kg IV 6-hourly** (add **ceftriaxone** if Gram-negative/negative stain or **sickle cell disease**: Salmonella); IV 2 weeks then oral to complete 4 weeks.
26	Physiological leg alignment by age?	**Genu varum** (bow legs) normal until about **2 years**; **genu valgum** (knock knees) common at **3–5 years**; then straightens.
27	Spinal examination findings and their significance?	**Sacral dimple/hair tuft/lipoma** = occult dysraphism; **rib hump on forward bend** = scoliosis; **gibbus** = TB spine (Pott's); sensory level/retention = cord compression.
28	What is nodding syndrome?	An epilepsy syndrome of **northern Uganda** (ages ~5–15): repeated **head-nodding** atonic seizures (often triggered by food or cold), cognitive decline, stunting; linked to **onchocerciasis** (UCG 2023 section 9.1.2).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 2-year-old with malaria is unconscious after a seizure 90 minutes ago. When firm pressure is applied to her sternum with a pencil, she pulls her arm away from a nail-bed stimulus but does not bring her hand to the sternum; she moans; she does not follow her mother's face. What is her Blantyre coma score?

A 1
B 2
C 3
D 4
E 5

2. A 7-month-old (7 kg) has fever and irritability. You plan an LP. Before proceeding, which finding would MOST make you defer the LP and start antibiotics first?

A Absent neck stiffness
B Temperature 39.5°C
C A bulging fontanelle while the baby is upright and calm
D Unequal pupils with intermittent extensor posturing
E Positive malaria RDT

3. A 6-year-old boy (20 kg) has had 5 days of progressive leg weakness that is now affecting his arms, beginning 2 weeks after a diarrhoeal illness. Reflexes are absent, plantars flexor, sensation and bladder normal. He is fully immunised. What is the MOST appropriate immediate step besides admission?

A Notify the case as acute flaccid paralysis, collect 2 stool samples 24 h apart, and monitor respiratory function closely
B Give IM vitamin B complex injections daily
C Discharge with physiotherapy exercises
D Start high-dose oral prednisolone
E Urgent spinal X-ray to look for TB

4. A 5-month-old is assessed for 'floppiness'. She smiles, follows faces and vocalises. She has marked head lag, lies in a frog-leg posture, has absent knee and biceps reflexes and fine twitching of the tongue. What is the MOST likely site of the problem? Advanced

A Cerebral cortex (central hypotonia)
B Basal ganglia
C Cerebellum
D Neuromuscular junction
E Anterior horn cells of the spinal cord

5. A 9-month-old is examined for developmental review. Which finding is ABNORMAL?

A Bilateral upgoing plantar responses
B Absent Moro reflex
C A persistent obligatory asymmetric tonic neck reflex
D Present parachute reflex
E Three beats of ankle clonus that then stop

6. A 7-year-old girl walks with a high-stepping gait on the right and her right foot slaps the ground. She cannot dorsiflex the right ankle; sensation is reduced over the dorsum of the foot; reflexes are normal elsewhere. Her mother says she had 'a quinine injection in the buttock' for malaria 6 weeks ago. What is the MOST likely diagnosis?

- **A** Right hemiplegic cerebral palsy
- **B** Guillain–Barré syndrome
- **C** Poliomyelitis
- **D** Sciatic (common peroneal division) nerve injection injury
- **E** Duchenne muscular dystrophy

7. A 4-year-old boy is examined for clumsiness and frequent falls. He gets up from the floor by turning prone and pushing his hands up his thighs; his calves are large and firm; knee jerks are reduced. Which investigation is MOST useful next?

- **A** EEG
- **B** Serum creatine kinase
- **C** CT brain
- **D** Lumbar puncture
- **E** Blood glucose

8. A 3-year-old with sickle cell disease suddenly cannot use her right arm and leg, and her mouth droops on the right. Right-sided reflexes are brisk with an upgoing plantar. Where is the lesion and what is the MOST likely cause?

- **A** Right cerebral hemisphere; meningitis
- **B** Spinal cord at C5; trauma
- **C** Right brachial plexus; birth injury
- **D** Cerebellum; malaria
- **E** Left cerebral hemisphere; ischaemic stroke

9. A 4-year-old (16 kg) has had fever for 2 days and refuses to walk. His right hip is held flexed, abducted and externally rotated, and he screams with any movement. T 39.2°C, WCC 18 × 10^9/L, ESR 65 mm/h. What is the MOST appropriate next step?

- **A** Urgent ultrasound-guided or surgical aspiration/drainage of the hip, blood culture, and IV cloxacillin 800 mg 6-hourly
- **B** Discharge with ibuprofen for transient synovitis
- **C** Skin traction and review in 1 week
- **D** Oral amoxicillin and review in 3 days
- **E** Wait for X-ray changes before treating

10. A 12-year-old boy from Gulu has had 6 months of back pain, weight loss and night sweats; now his legs are weak and stiff with brisk reflexes, upgoing plantars and reduced sensation below the umbilicus. There is a sharp angular prominence in his mid-back. What is the MOST likely diagnosis?

- **A** Scoliosis
- **B** Guillain–Barré syndrome
- **C** Spinal tuberculosis (Pott's disease) with cord compression
- **D** Nodding syndrome
- **E** Duchenne muscular dystrophy

11. You are examining a 3-year-old in an OSCE. Which sequence is MOST appropriate?

A Introduce yourself and gain consent, observe the child playing and walking, then examine tone, power and reflexes, leaving fundoscopy and any painful tests to the end

B Start with fundoscopy while the child is calm, then reflexes, then observation

C Test neck stiffness first in every child to exclude meningitis

D Undress the child completely and lie them flat before saying hello

E Test the plantar response first because it is the most important sign

12. A 3-year-old (14 kg) with fever and a malaria smear of 250,000/µL has a Blantyre score of 1, equal reactive pupils and no posturing. Fundoscopy shows patchy retinal whitening and white-centred haemorrhages. Glucose is 4.8 mmol/L. Which management is MOST appropriate? Advanced

A Oral artemether–lumefantrine and observation

B LP is contraindicated by the retinal findings; give no antibiotics

C IV quinine only

D Await CT before any treatment

E IV artesunate 42 mg at 0, 12 and 24 h then daily, plus ceftriaxone 1.4 g/day if meningitis cannot be excluded, with 2-hourly coma score and glucose

13. An 18-month-old girl has bowed legs. She walks well, is growing along the 50th centile, has no wrist widening or rib beading, and the bowing is symmetrical. What is the BEST management?

A Start high-dose vitamin D

B Refer for corrective osteotomy

C Reassure: physiological genu varum, and review growth and alignment in 6–12 months

D Apply long-leg casts

E Order a skeletal survey for abuse

14. A 10-year-old boy from Kitgum has 2 years of repeated episodes in which his head drops forward several times when he starts eating, with worsening school performance and poor growth. Neurological examination between episodes shows no focal signs. What is the MOST likely diagnosis?

A Febrile convulsions

B Nodding syndrome

C Guillain–Barré syndrome

D Cerebral palsy

E Breath-holding attacks

15. A 6-month-old is brought with a large head. Which combination of findings MOST strongly suggests raised intracranial pressure from hydrocephalus?

A Head circumference on the 97th centile since birth, both parents with large heads, normal development

B Flat anterior fontanelle and closed posterior fontanelle

C Frontal bossing with wide wrists and rib beading

D Head circumference crossing from the 50th to above the 99th centile, tense bulging fontanelle, split sutures and sunsetting eyes

E Plagiocephaly with a flat occiput on one side

Answers & explanations

1. B · 2

B is right: **motor 1** (withdraws, does not localise) + **verbal 1** (moan) + **eyes 0** (does not follow) = **2**, i.e. unrousable coma. A is wrong: it would require two components scoring 0. C is wrong: 3 would need localising pain (motor 2) or an appropriate cry (verbal 2). D and E are wrong: these indicate a nearly or fully conscious child.

2. D · Unequal pupils with intermittent extensor posturing

D is right: **unequal pupils and posturing** are WHO signs of **raised ICP** with herniation risk, so the LP should be deferred and **ceftriaxone** given immediately. A is wrong: neck stiffness is often absent in infants and is not a contraindication. B is wrong: fever is not a contraindication. C is wrong: a bulging fontanelle on its own suggests meningitis; WHO lists unequal pupils, posturing, limb paralysis and irregular breathing as the contraindications, so judge it with the senior, but it is not the most decisive here. E is wrong: parasitaemia does not exclude meningitis or contraindicate LP.

3. A · Notify the case as acute flaccid paralysis, collect 2 stool samples 24 h apart, and monitor respiratory function closely

A is right: ascending areflexic weakness after diarrhoea is **Guillain–Barré syndrome**, but every child <15 with AFP must be **notified with 2 stool samples** (polio surveillance), and the main danger is **respiratory failure**. B is wrong: IM injections are avoided in AFP and vitamins do not treat GBS. C is wrong: progressive weakness needs inpatient monitoring. D is wrong: steroids are not effective in GBS. E is wrong: there is no sensory level, back pain or gibbus to suggest cord compression.

4. E · Anterior horn cells of the spinal cord

E is right: an **alert** infant with severe weakness, **areflexia** and **tongue fasciculation** has **spinal muscular atrophy** (anterior horn cell disease). A is wrong: central hypotonia usually comes with poor alertness and preserved or brisk reflexes. B is wrong: basal ganglia lesions cause dyskinesia, not areflexic weakness. C is wrong: cerebellar lesions cause ataxia with mild hypotonia and normal power. D is wrong: neuromuscular junction disorders (infant botulism, myasthenia) cause fatigable weakness and ptosis, without fasciculation.

5. C · A persistent obligatory asymmetric tonic neck reflex

C is right: the **ATNR should disappear by 4–6 months**; persistence (especially obligatory) at 9 months suggests **cerebral palsy**. A is wrong: upgoing plantars are normal until about 12–18 months. B is wrong: the Moro reflex normally disappears by 3–6 months. D is wrong: the parachute reflex normally appears at 6–9 months. E is wrong: a few unsustained beats of clonus can be normal; sustained clonus (>3–5 beats) is abnormal.

6. D · Sciatic (common peroneal division) nerve injection injury

D is right: unilateral **foot drop** with sensory loss over the dorsum after a **buttock injection** points to **sciatic nerve injection injury** (peroneal division most vulnerable). A is wrong: CP is present from early life and causes spasticity, not a new LMN foot drop. B is wrong: GBS is bilateral, ascending and areflexic. C is wrong: polio causes no sensory loss. E is wrong: Duchenne causes symmetrical proximal weakness in boys.

7. B · Serum creatine kinase

B is right: **Gowers sign** with **calf pseudohypertrophy** in a boy suggests **Duchenne muscular dystrophy**; **CK** is markedly raised (often >10 times normal). A is wrong: there are no seizures. C is wrong: the findings point to muscle, not brain. D is wrong: CSF is not helpful in muscular dystrophy. E is wrong: glucose does not explain chronic proximal weakness.

8. E · Left cerebral hemisphere; ischaemic stroke

E is right: right **face, arm and leg** with **UMN signs** localise to the **left hemisphere**; in sickle cell disease the likely cause is **ischaemic stroke** (urgent transfusion per protocol, then imaging). A is wrong: a right-hemisphere lesion causes left-sided signs. B is wrong: a cord lesion spares the face and usually affects both sides below the level. C is wrong: a plexus injury causes LMN signs in one arm only, not the face and leg. D is wrong: cerebellar lesions cause ataxia, not hemiparesis with UMN signs.

9. A · Urgent ultrasound-guided or surgical aspiration/drainage of the hip, blood culture, and IV cloxacillin 800 mg 6-hourly

A is right: with all four **Kocher criteria** (fever >38.5°C, non-weight-bearing, ESR >40, WCC >12), **septic arthritis of the hip** is very likely; it needs urgent **drainage** and IV **cloxacillin 50 mg/kg** (16 × 50 = 800 mg) 6-hourly (UCG), adding ceftriaxone if sickle cell disease or Gram-negative. B is wrong: transient synovitis children are usually afebrile and well. C is wrong: delay risks avascular necrosis and joint destruction. D is wrong: oral amoxicillin is inadequate for septic arthritis. E is wrong: X-rays are normal early.

10. C · Spinal tuberculosis (Pott's disease) with cord compression

C is right: chronic back pain, constitutional symptoms, a **gibbus** and a **sensory level with UMN signs** indicate **spinal TB with cord compression**; urgent imaging, TB treatment and surgical review are needed. A is wrong: scoliosis is a lateral curve without a gibbus or cord signs. B is wrong: GBS causes areflexic weakness without a sensory level or gibbus. D is wrong: nodding syndrome causes head-nodding seizures, not cord compression. E is wrong: Duchenne causes proximal weakness with reduced reflexes and no sensory signs.

11. A · Introduce yourself and gain consent, observe the child playing and walking, then examine tone, power and reflexes, leaving fundoscopy and any painful tests to the end

A is right: in young children, **observation and play come first**, hands-on examination next, and **unpleasant tests last**, after consent. B is wrong: fundoscopy early upsets the child and loses cooperation. C is wrong: meningism is tested when relevant and is unpleasant; it is not the first step in every child (and never before excluding neck injury in trauma). D is wrong: introductions, consent and dignity come first; full undressing is unnecessary. E is wrong: the plantar response is unpleasant and less informative than observation in a toddler.

12. E · IV artesunate 42 mg at 0, 12 and 24 h then daily, plus ceftriaxone 1.4 g/day if meningitis cannot be excluded, with 2-hourly coma score and glucose

E is right: a Blantyre score ≤2 with parasitaemia and **malarial retinopathy** = **cerebral malaria**; give **IV artesunate 3 mg/kg** (14 × 3 = 42 mg) and, unless an LP excludes it, cover meningitis with **ceftriaxone 100 mg/kg/day** (1.4 g); monitor coma score and glucose. A is wrong: oral treatment is unsafe and inadequate in severe malaria. B is wrong: malarial retinopathy is not a contraindication to LP and meningitis still needs cover. C is wrong: artesunate is superior to quinine. D is wrong: treatment must not wait for imaging.

13. C · Reassure: physiological genu varum, and review growth and alignment in 6–12 months

C is right: **symmetrical bowing up to about 2 years** in a well-growing child without signs of rickets is **physiological genu varum**, which corrects spontaneously; reassure and review. A is wrong: there are no clinical features of rickets (check diet and X-ray only if signs appear). B and D are wrong: surgery and casting are not indicated for physiological bowing. E is wrong: there is nothing to suggest abuse.

14. B · Nodding syndrome

B is right: repeated **head-nodding** (atonic) seizures triggered by **food**, cognitive decline and stunting in a child from **northern Uganda** is typical of **nodding syndrome** (UCG 2023 section 9.1.2; associated with onchocerciasis). A is wrong: he is 10 years old and the episodes are afebrile. C is wrong: GBS causes acute flaccid weakness. D is wrong: CP is a static motor disorder present from early life, not new seizures with decline. E is wrong: breath-holding attacks occur in toddlers, triggered by crying.

15. D · Head circumference crossing from the 50th to above the 99th centile, tense bulging fontanelle, split sutures and sunsetting eyes

D is right: an HC **crossing centiles upward** with a **tense fontanelle, split sutures and sunsetting eyes** indicates **progressive hydrocephalus** with raised pressure. A is wrong: a large but stable head in a well child with large-headed parents is benign familial macrocephaly. B is wrong: a flat fontanelle and a closed posterior fontanelle at 6 months are normal. C is wrong: frontal bossing with wrist widening and a rosary suggests rickets. E is wrong: positional plagiocephaly is a moulding deformity without raised pressure.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The unconscious child on the malaria ward

Akello, a 3-year-old girl (12 kg) from Masaka, was admitted to Mengo with 2 days of fever and 3 generalised seizures. An hour after her last seizure she has not woken up. You are asked to assess her neurologically.

1. What do you do before the detailed neurological examination?

> **ABCD**: airway (recovery position, suction), breathing (RR, SpO_2, oxygen), circulation (CRT, pulse), **AVPU**, **glucose** (treat <3.0 mmol/L with 10% dextrose 5 mL/kg = 60 mL), check for ongoing or subtle seizures, temperature.

2. How will you score her conscious level?

> **Blantyre coma scale**: motor (sternal/supraorbital pressure → localises = 2; nail-bed pressure → withdraws = 1; none = 0), verbal (appropriate cry 2, moan 1, none 0), eyes (follows mother's face 1, not 0). She withdraws, moans and does not follow: **M1 V1 E0 = 2** (unrousable coma).

3. What else will you examine, and why?

> **Pupils** and **posture** (raised ICP), **fontanelle** (closed at 3 y), **neck stiffness/Kernig** (meningitis), **fundoscopy** (malarial retinopathy, papilloedema), tone, reflexes, plantars and limb movements for **focal signs**, breathing pattern, pallor, jaundice, spleen, signs of dehydration and **acidotic breathing**.

4. Findings: pupils equal and reactive, no posturing, regular breathing, no neck stiffness, malarial retinopathy present, symmetrical reflexes, flexor plantars. Smear 180,000/µL; Hb 7.2; glucose 4.6. What is your summary and plan?

> **Cerebral malaria** (BCS 2, retinopathy, repeated seizures); no signs of raised ICP, so **LP** is reasonable to exclude meningitis (or give ceftriaxone if LP is delayed). **IV artesunate 36 mg** (3 mg/kg) at 0, 12, 24 h then daily; seizure management; maintenance fluids (avoid overload); NG feeds if still unconscious after 24 h; 2-hourly BCS and glucose.

5. Six hours later her left pupil is larger and sluggish, she has extensor posturing and irregular breathing. What do you do?

> **Signs of raised ICP**: call the senior/anaesthetist, airway protection (consider intubation where possible), head-up 30° midline, avoid hypotonic fluids, treat seizures and fever, check glucose and Hb; **no LP**; give ceftriaxone if not already started; ICU referral. Routine mannitol/steroids are not recommended in cerebral malaria (WHO); specialist decision.

6. She recovers. What neurological follow-up is needed?

> Examine before discharge (gait, ataxia, tone, vision, hearing, speech, behaviour); arrange review for **neurological sequelae** (epilepsy, hemiparesis, ataxia, cognitive and behaviour problems), **post-discharge malaria chemoprevention** (monthly DP for 3 months, UCG 2023), and a development check at follow-up.

Take-away: In coma, score the components, look at the pupils, posture and fundi, and let the examination decide whether an LP is safe.

Case 2: The child who cannot stand

Mugisha, a 5-year-old boy (18 kg) from Mbarara, has been unable to stand for 2 days. A week earlier he had diarrhoea. His mother says his legs 'became weak like water' and now he struggles to lift his arms. His immunisation card shows all OPV and IPV doses.

1. What features of the history help you localise the problem?

> Onset and progression (days, ascending), preceding infection (GBS), pain, fever at onset (polio), symmetry, **bladder and bowel** (cord), **back pain** (cord compression), sensory symptoms, breathing or swallowing difficulty (respiratory/bulbar involvement), **injections**, toxins/poisoning, immunisation.

2. Outline your neurological examination.

> ABC with **respiratory assessment** (RR, SpO_2, single-breath count, cough strength); cranial nerves (facial weakness, swallowing); inspection; **tone**; **power** (MRC, proximal vs distal); **reflexes** and plantars; **sensation** and a **sensory level**; bladder (palpable?); spine (tenderness, gibbus, midline markers); gait if possible.

3. Findings: symmetrical flaccid weakness (legs 2/5, arms 4/5), absent reflexes, flexor plantars, no sensory level, bladder normal, mild bilateral facial weakness, single-breath count 14. What is your diagnosis?

> **LMN pattern**, ascending, symmetrical, areflexic, with facial involvement and no cord level: **Guillain–Barré syndrome**, possibly triggered by *Campylobacter* diarrhoea.

4. What must you do immediately in Uganda?

> Treat as **acute flaccid paralysis**: notify the district surveillance focal person, collect **2 stool samples 24 h apart within 14 days** (cold chain), **avoid IM injections**. Admit for close monitoring of breathing and swallowing (4-hourly single-breath count/SpO_2), NG feeds if swallowing unsafe, DVT/pressure care, pain control.

5. When would you escalate, and what specific treatment exists?

> Escalate to HDU/ICU if single-breath count falls, SpO_2 drops, weak cough, bulbar weakness or autonomic instability. **IVIG** (2 g/kg over 2–5 days) or plasma exchange where available (mostly private/national referral); steroids are not effective. Physiotherapy throughout; most children recover over weeks to months.

Take-away: Localise first (UMN, LMN, muscle, cord), then remember that every AFP in a child under 15 is notified and sampled for polio.

Case 3: The limping child with fever

Nakato, a 6-year-old girl (20 kg) with sickle cell anaemia, presents to the paediatric emergency unit with 3 days of fever and pain in the left leg; she now refuses to walk. She has had painful crises before.

1. What is your differential diagnosis?

> **Vaso-occlusive crisis**, **osteomyelitis** (S. aureus, **Salmonella** in sickle cell), **septic arthritis** (hip or knee), **pyomyositis**, avascular necrosis of the femoral head, fracture, and less likely leukaemia.

2. How do you examine her?

> ABC, vitals, pallor, jaundice, hydration; **gait** (refuses), observe the leg position; **look–feel–move**: swelling, redness, warmth, point **bony tenderness** at the metaphysis (osteomyelitis) vs **joint** tenderness, effusion and painful restricted movement (septic arthritis), deep muscle swelling (pyomyositis); hip examination (flexed, abducted, externally rotated = septic hip); examine joints above and below; neurovascular check; spine; other bones.

3. Findings: T 39°C, swollen warm left knee with effusion, severely painful on any movement; no bony point tenderness. WCC 22, CRP high. What do you do?

> Likely **septic arthritis of the left knee**: analgesia (paracetamol 15 mg/kg, opioid if needed), splint, blood culture, **joint aspiration** (Gram stain, culture), hydration, and **IV cloxacillin 50 mg/kg (1 g) 6-hourly plus ceftriaxone** (sickle cell: cover Salmonella) per UCG; orthopaedic referral for washout.

4. How long will antibiotics continue and what will you monitor?

> IV for about **2 weeks**, then oral (e.g. per culture) to complete **4 weeks** (longer if osteomyelitis); monitor fever, CRP, joint swelling and range, Hb (sickle crisis, aplastic crisis), pain; physiotherapy to restore movement.

5. What long-term advice is important?

> Sickle cell care: **hydroxyurea**, folic acid, malaria prevention (bed nets, chemoprophylaxis per UCG), penicillin prophylaxis in young children, **PCV** and other vaccines, early review for fever or bone pain; return immediately if the joint becomes painful again.

Take-away: A febrile child who will not move a joint has septic arthritis until aspiration proves otherwise; in sickle cell disease cover Salmonella.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.